Time-Varying Treatments and Marginal Structural Models

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Proposenity Score: A Review for Single Time Point

- Defined for a binary treatment to be $P(A = 1|X) = \pi(X)$
- Under the assumption of no unmeasured confounding and consistency, the propensity score is a balancing score
- A balancing score $b(X)$ is any function of the covariates X such that A is independent of X given $b(X)$
- This means the distribution of the covariates is the same in the treatment and control group at the same level of $b(X)$
- Note that $P(A = 1|X) = P(A = 1|W)$ –> show this

Proposenity Score: Multiple Time Points

- Note that it is difficult (impossible?) to talk about a single propensity score when there are multiple time points
- If we fix $\overline{a}$ then we will define the propensity score (for that sequence of treatments) as $P(\overline{A} = \overline{a}|W)$ where W denotes the set of all potential outcomes
- Under consistency and sequential ignorability one can show that

$$P(\overline{A} = \overline{a}|W) = P(A_0 = a_0|\underline{L_0}) \times \underline{P}(A_1 = a_1|A_0 = a_0, \overline{L}_1) \times ... \times P(A_M = a_m|\overline{A}_{M-1}, \overline{L}_M)$$

- That is, the propensity score (for sequence of treatments $\overline{a}$) is the product that at each timepoint a person takes treatment a_j given all the information previously collected on them

Model for Probability of Receiving Treatment

	Estimate	Std. Error	z value	Pr(>|z|)
(Intercept)	2.96	0.38	7.80	0
logCD4	-0.46	0.06	-7.50	0
R	-1.10	0.12	-9.25	0

Inverse Probability Weighting

- Assume that we have a sample of observed data $(\overline{L}_i, \overline{A}_i, Y_i)$ for $i = 1, \ldots, n$
- We can derive a consistent estimator for $E(Y^{\overline{a}})$ by using

$$n^{-1} \sum_{i=1}^{n} \frac{I(\overline{A}_i = \overline{a})}{\prod_{j=0}^{M} P_{A_j|\overline{A}_{j-1}, \overline{L}_j}(a_j | \overline{a}_{j-1}, \overline{L}_{ij})} Y_i \tag{1}$$

- SHOW this is consistent

Inverse Probability Weighting for Sequence (0, 1, 1, 1)

```
dataHW3_long <-  mutate(dataHW3_long, pred_trt = predict(m3, r
    pred_trt_rule = ifelse(time > 0, pred_trt, 1-pred_trt),
    trt_rule = ifelse((time == 0 & A == 0)|(time > 0 & A ==1),
den_wt <- tapply(dataHW3_long$pred_trt_rule,
    dataHW3_long$ID, prod, na.rm = TRUE)
num_wt <- tapply(dataHW3_long$trt_rule,
    dataHW3_long$ID, prod, na.rm = TRUE)
sum(num_wt)
```

```
[1] 68
```

```
round(weighted.mean(dataHW3$CD44, w = num_wt/den_wt))
```

```
[1] 525
```

Limitations of Inverse Probability Weighting

- In order for this method to be useful, we cannot have too many decision points M or many decision options at each point
- Otherwise, the number of individuals such that $I(\overline{A}_i = \overline{a})$ becomes so small as to give very unstable estimators
- For example, if a binary treatment were randomly assigned with 50% probability at 4 decision points, the number of subjects such that $I(\overline{A}_i = \overline{a})$ would be $1/16$ the total sample size
- Typically, we avoid this problem of small sample sizes through modeling. For example, we use additive regression models rather than fit separate models for each covariate combination

Modeling potential outcomes

- Consider the example in the homework - binary treatment at 4 decision points (16 different treatment combinations)
- Could (in theory) estimate a different expected outcome for each treatment sequence
- BUT we might have more precision if we could pool some of these treatment sequences together

Marginal Structural Model

- Let's break down the name
- Structural = having to do with potential outcomes
- Marginal = not covariate adjusted
- Overall, this is a proposed model for (usually) the mean of potential outcomes based on combining different treatment sequences together
- Example: $E(Y^{\overline{a}}) = \beta_0 + \beta_1 cum(\overline{a})$ where $cum(\overline{a}) = \sum_{l=0}^{M} a_l$. Instead of estimating
- Other possibilities?
- Note that at this point I have not discussed estimation

Estimating a Marginal Structural Model

- MSM are models for potential outcomes (which we may not observe)
- If we want to use observed data to estimate (functionals of) potential outcomes need to worry about confounding
- In other words, we cannot consistently estimate β_0 and β_1 by fitting a regression model with Y_i as the outcome and $cum(\overline{A}_i)$ as the predictor
- Instead we need to use weighted regression with weights $\frac{1}{\prod_{j=0}^{M} P_{A_j|\overline{A}_{j-1},\overline{L}_j}(A_{ij}|\overline{A}_{i,j-1},\overline{L}_{ij})}$
- This is one over the probability that a particular subject would have followed their observed treatment history

Marginal Structural Model for Treatment Initiation

- In the HIV example, want to compare different different times for initiating (and then sustaining) treatment
- Compare the regimes (0, 0, 0, 0); (0, 0, 0, 1); (0, 0, 1, 1); (0, 1, 1, 1); (1, 1, 1, 1);
- Note that there are 5 different regimes here so a saturated model would have 5 parameters
- Consider time period when treatment was initiated as predictor with additional quadratic term (3 parameter model including intercept); never initiators have time set to 4

Marginal Structural Model - Denominator Weights

```
dataHW3_long <-   mutate(dataHW3_long,
    pred_trt = predict(m3, newdata = dataHW3_long, type = "res
    pred_trt_received = (pred_trt^A)*(1-pred_trt)^(1-A))
# calculate denominator weights
dataHW3$den_wt <- tapply(dataHW3_long$pred_trt_received,
    dataHW3_long$ID, prod, na.rm = TRUE)
```

Marginal Structural Model - Initiation Time

```
dataHW3 <- mutate(dataHW3,
    init_time =     NA,
    init_time = ifelse(A0 == 1 & A1 == 1 & A2 == 1 & A3 == 1,
    init_time = ifelse(A0 == 0 & A1 == 1 & A2 == 1 & A3 == 1,
    init_time = ifelse(A0 == 0 & A1 == 0 & A2 == 1 & A3 == 1,
    init_time = ifelse(A0 == 0 & A1 == 0 & A2 == 0 & A3 == 1,
    init_time = ifelse(A0 == 0 & A1 == 0 & A2 == 0 & A3 == 0,
```

Estimate Marginal Structural Model

```r
MSM_CD4 <- lm(CD44 ~   init_time + I(init_time^2), data = data
round(summary(MSM_CD4)$coefficients, digits = 2)
```

```
               Estimate Std. Error t value Pr(>|t|)
(Intercept)      406.98      44.42    9.16     0.00
init_time        131.39      49.19    2.67     0.01
I(init_time^2)   -34.76      11.54   -3.01     0.00
```

Estimate Marginal Structural Model

```r
newdatatime <-   data.frame(init_time = 4:0)
predCD4 <- data.frame(CD4 = predict(MSM_CD4, newdatatime))
rownames(predCD4) <- c("(0, 0, 0, 0)",
    "(0, 0, 0, 1)",
    "(0, 0, 1, 1)",
    "(0, 1, 1, 1)",
    "(1, 1, 1, 1)")
print(round(predCD4))
```

```
             CD4
(0, 0, 0, 0) 376
(0, 0, 0, 1) 488
(0, 0, 1, 1) 531
(0, 1, 1, 1) 504
(1, 1, 1, 1) 407
```

Obtain Standard Errors Using Bootstrap

	CD4	SE
(0, 0, 0, 0)	376	42
(0, 0, 0, 1)	488	25
(0, 0, 1, 1)	531	25
(0, 1, 1, 1)	504	21
(1, 1, 1, 1)	407	38

Comparison Between G-computation and MSM